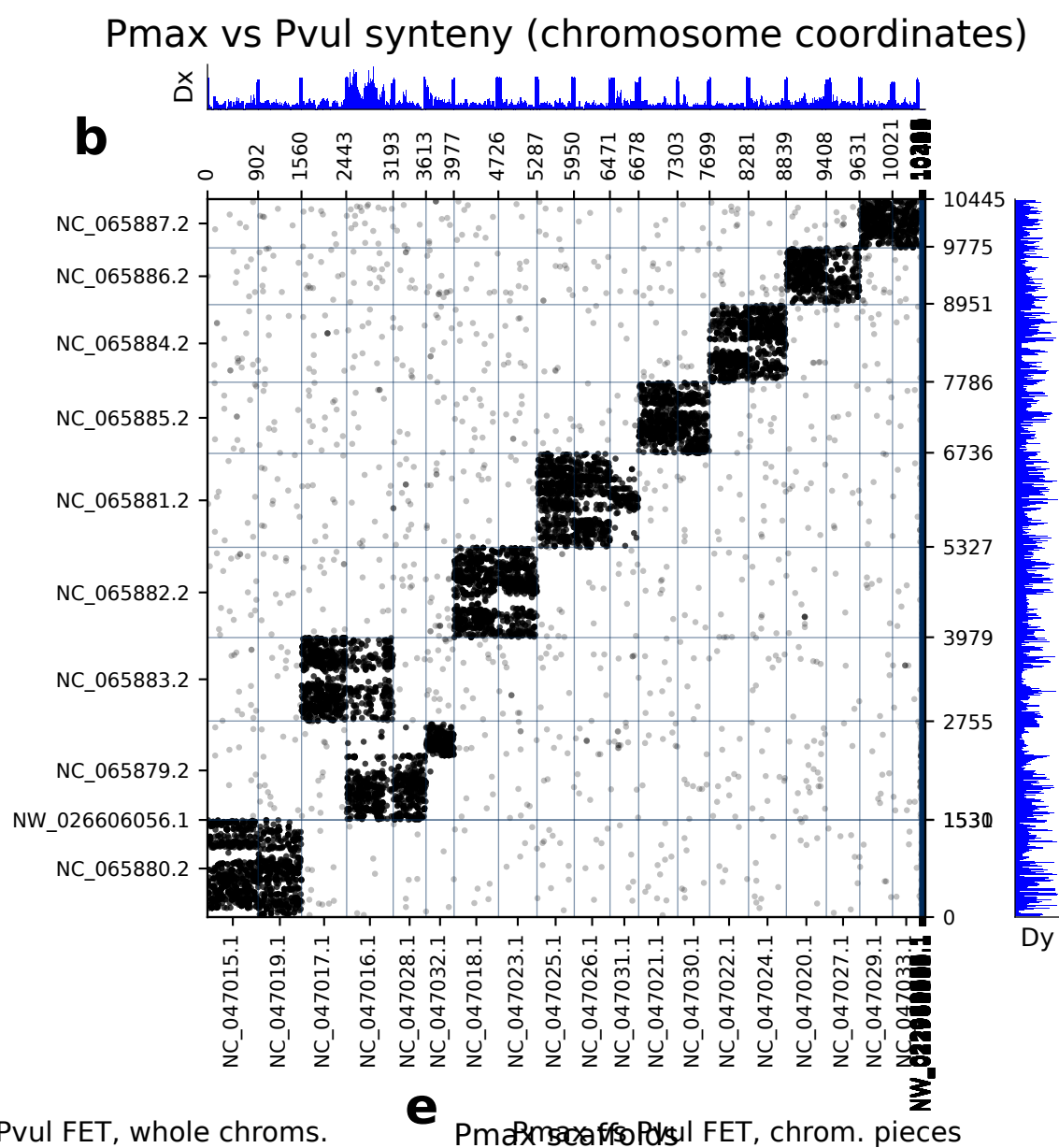
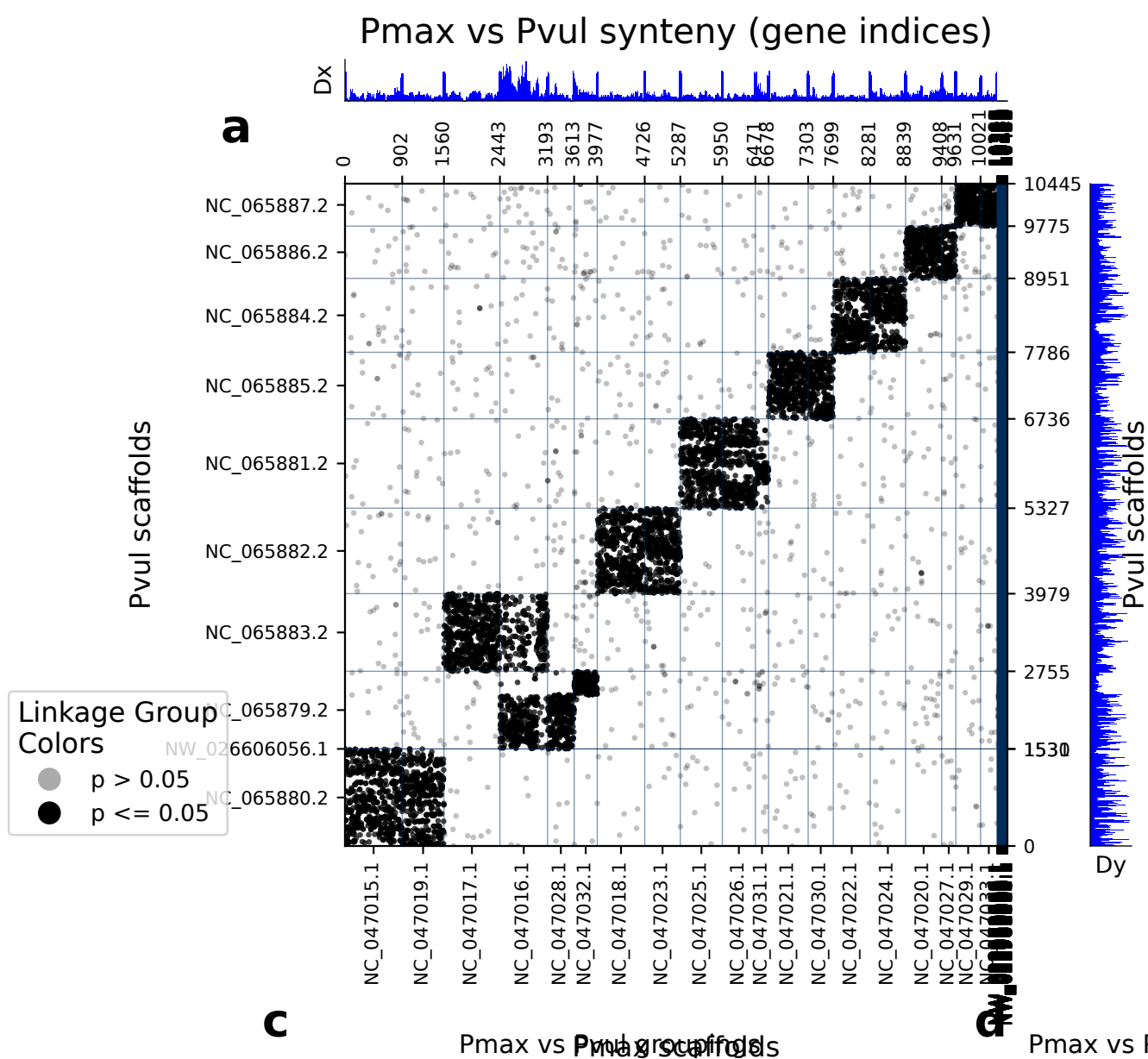


diamond: 10446 orthologs on 175 Pmax scaffolds and 10 Pvul scaffolds.



P_{max} vs P_{max} scaffolds

Pmax vs Pvul FET, whole chroms.

	Pmax scaf	Pvul scaf	p (FET)	Count
0	NC_047015.1	NC_065880.2	0.00e+00	813
1	NC_047017.1	NC_065883.2	0.00e+00	791
2	NC_047018.1	NC_065882.2	0.00e+00	676
3	NC_047025.1	NC_065881.2	0.00e+00	600
4	NC_047019.1	NC_065880.2	0.00e+00	589
5	NC_047021.1	NC_065885.2	0.00e+00	560
6	NC_047022.1	NC_065884.2	0.00e+00	532
7	NC_047023.1	NC_065882.2	0.00e+00	501
8	NC_047020.1	NC_065886.2	0.00e+00	493
9	NC_047024.1	NC_065884.2	0.00e+00	488
10	NC_047026.1	NC_065881.2	0.00e+00	479
11	NC_047016.1	NC_065879.2	1.27e-178	391
12	NC_047028.1	NC_065879.2	5.80e-288	362
13	NC_047030.1	NC_065885.2	0.00e+00	355
14	NC_047029.1	NC_065887.2	0.00e+00	339
15	NC_047016.1	NC_065883.2	4.71e-85	289
16	NC_047032.1	NC_065879.2	1.21e-195	283
17	NC_047033.1	NC_065887.2	2.08e-233	222
18	NC_047027.1	NC_065886.2	1.50e-168	185
19	NC_047031.1	NC_065881.2	5.62e-103	165

Pmax scaffolds Pul FET, chrom. pieces

	Pmax scaf	Pvul scaf	p (FET)	Count
0	NC_047015.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	813
1	NC_047017.1 : 0-end	NC_065883.2 : 0-end	0.00e+00	791
2	NC_047018.1 : 0-end	NC_065882.2 : 0-end	0.00e+00	676
3	NC_047025.1 : 0-end	NC_065881.2 : 0-end	0.00e+00	600
4	NC_047019.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	589
5	NC_047021.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	560
6	NC_047022.1 : 0-end	NC_065884.2 : 0-end	0.00e+00	532
7	NC_047023.1 : 0-end	NC_065882.2 : 0-end	0.00e+00	501
8	NC_047020.1 : 0-end	NC_065886.2 : 0-end	0.00e+00	493
9	NC_047024.1 : 0-end	NC_065884.2 : 0-end	0.00e+00	488
10	NC_047026.1 : 0-end	NC_065881.2 : 0-end	0.00e+00	479
11	NC_047016.1 : 0-end	NC_065879.2 : 0-end	1.27e-178	391
12	NC_047028.1 : 0-end	NC_065879.2 : 0-end	5.80e-288	362
13	NC_047030.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	355
14	NC_047029.1 : 0-end	NC_065887.2 : 0-end	0.00e+00	339
15	NC_047016.1 : 0-end	NC_065883.2 : 0-end	4.71e-85	289
16	NC_047032.1 : 0-end	NC_065879.2 : 0-end	1.21e-195	283
17	NC_047033.1 : 0-end	NC_065887.2 : 0-end	2.08e-233	222
18	NC_047027.1 : 0-end	NC_065886.2 : 0-end	1.50e-168	185
19	NC_047031.1 : 0-end	NC_065881.2 : 0-end	5.62e-103	165

(a) depicts the chromosomes/scaffolds of Pmax (x-axis) plotted against the chromosomes/scaffolds of Pvul (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10446 orthologs found between 175 Pmax scaffolds and 10 Pvul scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.